User guide: `YukawaVPS`

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1 Introduction

This repository accompanies [1] and provides the numerical implementation and tabulated data for the treatment of long-range scattering in the Yukawa potential. The code uses the variable-phase method to compute partial-wave phase shifts, Sommerfeld factors, and quantities derived from the irregular solution. The repository also provides precomputed elastic, transfer, and viscosity cross sections, together with partial-wave convergence information.

The purpose of this guide is to describe the structure of the repository and provide the information needed to use the numerical implementation and accompanying datasets. It is organized as follows. Section 2 introduces the Yukawa scattering problem and the dimensionless parameters used throughout the numerical implementation, and summarizes the scattering observables provided by the repository. Section 3 describes the two Mathematica notebooks, including their numerical modules and example workflows. Section 4 documents the structure of the tabulated datasets, including the parameter grids, file organization, data formats, import routines, and interpolation. Section 5 discusses proper attribution and licensing.

2 Physical setup

This repository focuses on the numerical treatment of scattering in the Yukawa potential,

$$V^Y(r) = -\frac{\alpha}{r} e^{-m_{\rm med} r}, \tag{2.1}$$

where we work in natural units, $\hbar = c = 1$, here and throughout the repository. Above, α is the coupling strength, with $\alpha > 0$ ($\alpha < 0$) corresponding to an attractive (repulsive) potential, labeled `A` (`R`) throughout the repository, and $m_{\rm med}$ is the mediator mass. The two-body system is characterized by the reduced mass μ. In the center-of-mass frame, the relative momentum is $k = \mu v_{\rm rel}$, where $v_{\rm rel}$ is the relative velocity.

2.1 Dimensionless parameters

We introduce the dimensionless variables

$$\zeta \equiv \frac{\mu\alpha}{k} = \frac{\alpha}{v_{\rm rel}}, \qquad \xi \equiv \frac{\mu|\alpha|}{m_{\rm med}}. \tag{2.2}$$

Together with the dimensionless radial coordinate $x = kr$, the scattering problem can be expressed entirely in terms of ξ, ζ, and the partial-wave index ℓ. The numerical results presented in this repository are therefore tabulated on a grid in these variables.

The quantities provided in the repository include the partial-wave phase shifts and Sommerfeld factors, as well as the information required to construct the elastic, transfer, and viscosity cross sections. The definitions and detailed derivation of these quantities are given in [1].

2.2 Scattering observables

Having computed the partial-wave phase shifts $\theta_\ell(k)$ and Sommerfeld factors $S_\ell(k)$, we construct the elastic, transfer, and viscosity cross sections,

$$\sigma_{\mathrm{elas}}, \quad \sigma_{\mathrm{T}}, \quad \sigma_{\mathrm{V}},$$

from their partial-wave contributions. The tabulated cross sections provided with the repository assume distinguishable particles. In the partial-wave expressions, this corresponds to setting the symmetry factors $\mathscr{C}_\ell = 1$ for all ℓ. For the definition of these factors and the general expressions applicable to identical particles, see [1]. The phase-shift data can also be used to calculate cross sections for other choices of symmetry factors. For applications involving inelastic scattering, the repository additionally provides data obtained from the irregular solution, specifically $T(k; k_\diamond)$, which is used in the unitarization prescription described in [2–5].

3 Software and notebooks

3.1 Requirements

No external packages are required for the numerical calculations. The `MaTeX` package [6] is used for plotting and is recommended for reproducing the included examples as discussed below. All tabulated data are provided as plain-text `.txt` files and can therefore also be imported using other programming languages.

The code was developed and tested using `Mathematica` 13.3 and 14.3 on macOS and Linux. The tested systems include an Apple M5 Pro MacBook Pro with 48 GB of RAM running macOS Tahoe 26.6.2 (`Mathematica` 14.3), an Apple M2 Pro MacBook Pro with 32 GB of RAM running macOS Tahoe 26.6.2 (`Mathematica` 13.3), and a Dell Precision 5470 with a 12th Gen Intel Core i7-12800H processor and 32 GB of RAM running Ubuntu 24.04.4 LTS (`Mathematica` 14.3).

3.2 `YukawaVPS.nb`

`YukawaVPS.nb` is the main notebook of the repository. It is organized into three main chapters: `General Setup`, `Code`, and `Example evaluations`.

`General Setup`**:** Contains the notebook initialization, utility functions, and plotting settings used throughout the notebook.

`Code`**:** Contains the main numerical implementation, including the variable-phase calculation and routines for computing phase shifts, Sommerfeld factors, and the irregular-solution quantities. The `Code` chapter is further divided into three sections:

- `Variable phase shift setup`: This section contains the definitions and numerical setup required for the variable-phase calculation. This includes the Riccati–Bessel functions, the dimensionless Yukawa potential, the Coulomb-limit results

used for comparison, and the conversion between physical and dimensionless parameters. It also defines the integration boundaries and initial conditions used in the variable-phase equations.

The main module is `VPSSolve`, which solves the variable-phase equations and returns the phase shift and Sommerfeld factor for a given $(|\zeta|, \xi, \ell)$ for an attractive or repulsive potential, specified through the `sign` parameter. The module automatically determines the integration range and tests the convergence of the phase shift. If the calculation does not converge at the initial working precision, the working precision is increased and the calculation is repeated. A detailed discussion of the notation and formulas can be found in [1].

We also provide the `PhysVPSSolve` module, which performs the same calculation using the physical parameters $(m_{\rm DM}, m_{\rm med}, v_{\rm rel}, \alpha)$ as input, converting them to the corresponding dimensionless parameters before calling `VPSSolve`. `PhysVPSSolve` assumes equal masses, i.e., $\mu = m_{\rm DM}/2$.

- `Irregular solution computation`: This section contains the modules used to compute the irregular solution and the quantities derived from it. The calculation is restricted to the s-wave, $\ell = 0$.

  The main module is `IrregSolve`, which first determines the phase shift and Sommerfeld factor using `VPSSolve` and then integrates the variable-phase equations backward from the large-x boundary to construct the irregular solution.

  The `LinCoeffCalc` module extracts the s-wave linear coefficient of the irregular solution, denoted $\tilde{g}_0^{(1)}(k)/k$, using the method established in [1].

  The `TCompute` module uses these coefficients to compute the dimensionless function $T(k; k_\diamond)$. The `TZetaScan` module performs this calculation over a specified range of ζ for $\zeta_\diamond = 1$ and fixed ξ.

- `Parameter scan and data handling`: This section contains the modules used to perform parameter scans and to export, import, and interpolate the resulting data.

  The `dataOutput` module performs the multidimensional parameter scans used to generate the tabulated phase shifts and Sommerfeld factors. For a fixed partial wave ℓ, it scans over specified ranges of ξ and $|\zeta|$ with logarithmic spacing and evaluates `VPSSolve` at each point. The region $10\,|\zeta| < \xi$ is excluded from the scan. As before, the `sign` parameter specifies whether the potential is attractive or repulsive. The module returns the phase shifts and Sommerfeld factors as separate arrays.

  The `exportData` and `exportDataGrid` modules export the calculated results and corresponding parameter grid. The `importGrid`, `importPhaseShiftData`, `importSommerfeldData`, and `importTData` modules facilitate the import of the tabulated data, while `makeInterp` constructs interpolation functions over the parameter grid. The import modules assume the data directory `../data/` by default, which can be changed through the `"OutputDirectory"` option.

`Example evaluations:` Provides examples illustrating the main workflows of the notebook. These include:

- an example Sommerfeld-factor calculation;
- an example parameter scan;
- an example $T(k; k_\diamond)$ calculation;
- an example of importing tabulated phase shifts;
- an example of importing tabulated Sommerfeld factors;
- an example of importing tabulated $T(k; k_\diamond)$ data; and
- an example of constructing an interpolation function for the Sommerfeld factors.

By default, the example evaluations are commented out and must be uncommented before they can be evaluated. The `General Setup` and `Code` chapters should be run in their entirety to ensure that the examples perform as expected. Note that some of the example evaluations make use of `ParallelTable` or involve the manipulation of large data files. Both can be time- and resource-intensive.

3.3 `CrossSectionsVPS.nb`

`CrossSectionsVPS.nb` is the complementary notebook for computing cross sections relevant for self-interacting dark matter applications. It is similarly organized into three main chapters: `General Setup`, `Code`, and `Example evaluations`.

`General Setup:` Identical to that of `YukawaVPS.nb`.

`Code:` Contains the numerical modules used to compute and process the scattering cross sections. The code is organized into sections for parameter setup, cross-section calculations, grid-based calculations, and data handling, as described below.

- `Setup`: Defines the numerical precision and parameter-conversion modules used throughout the notebook.
- `Normalized scattering cross sections`: Provides modules for computing the elastic, transfer, and viscosity cross sections from the partial-wave phase shifts, normalized to the s-wave unitarity cross section, $\sigma^{\mathrm{U}}_{\ell=0} = 4\pi/k^2$. The tabulated cross sections included with the repository assume distinguishable particles, corresponding to $\mathscr{C}_\ell = 1$ for all partial waves. The phase-shift data can also be used to calculate cross sections for other choices of symmetry factors, using the appropriate general expressions given in [1].

- `Grid cross-section calculations`: Includes modules to apply the cross-section calculations over the full $(\xi, |\zeta|)$ parameter grid using the tabulated phase shifts. The partial-wave sums include partial waves up to $\ell = 100$, 99, and 98 for the elastic, transfer, and viscosity cross sections, respectively. In addition, the modules determine $\ell_{\max}$ using `convCalc`, which applies the geometric-tail estimate of the relative truncation error given in [1].

- `Cross-section computation and data handling`: Contains modules for importing the parameter grid and tabulated phase-shift data, computing and exporting the cross sections and corresponding $\ell_{\max}$ values (`exportCSLMAXData`), importing precomputed cross-section and $\ell_{\max}$ data (`importCSLMAXData`), and constructing interpolation functions over the parameter grid. As in `YukawaVPS.nb`, the default data directory can be changed through the `"OutputDirectory"` option.

`Example evaluations:` Provides examples illustrating the following workflows:

- computation and export of cross sections from phase-shift data;

- import of tabulated cross sections and construction of interpolation functions.

As above, the example evaluations are commented out by default and require the `General Setup` and `Code` chapters to be run before evaluation. Some examples use `ParallelTable` or involve the manipulation of large data files and may therefore be time- and resource-intensive.

4 Data structure and access

The results of the numerical calculations are provided as tabulated datasets for direct use in subsequent calculations. The following sections describe the parameter grids, data format, and quantities contained in the datasets.

4.1 Grid specifications

The tabulated data cover the ranges,

$$\begin{aligned} \log_{10}|\zeta| &\in [-4,\ 7], \\ \log_{10}\xi &\in [-6,\ 5], \qquad \text{excluding} \quad 10\,|\zeta| < \xi, \\ \ell &\in [0, 100], \end{aligned} \tag{4.1}$$

with a resolution set by

$$\Delta \log_{10}\xi = \Delta \log_{10}|\zeta| = 10^{-2}. \tag{4.2}$$

The region $10\,|\zeta| < \xi$ is excluded from the scan as it approaches the Coulomb limit. Here, the long-range nature of the potential requires contributions from an increasingly large

number of partial waves and therefore an increased computational cost. For this reason, the specified parameter range, together with the inclusion of the first 101 partial waves ($\ell = 0, \ldots, 100$), is a practical choice for the tabulated dataset rather than a limitation of the underlying numerical implementation.

4.2 Tabulated quantities

The repository provides tabulated results obtained using the variable-phase method developed in the accompanying paper [1]. These data allow the resulting scattering quantities to be used without repeating the numerical calculation. For both attractive and repulsive Yukawa potentials, we provide the partial-wave phase shifts $\theta_\ell(k)$ and Sommerfeld factors $S_\ell(k)$ for $\ell = 0, \ldots, 100$. These constitute the main outputs of the variable-phase calculation. Note that the phase shifts are given in the interval $(-\pi, \pi]$ and are therefore not continuous across the grid.

In addition, two further calculations are performed. First, for applications involving inelastic scattering, we provide quantities obtained from the irregular solution for the attractive Yukawa potential. The quantity of interest is the dimensionless function $T(k; k_\diamond)$, which is tabulated on the same $(\xi, |\zeta|)$ grid for fixed $\zeta_\diamond = 1$ and for the s-wave, $\ell = 0$ only. These data are provided for the attractive case only, as the unitarization prescription of interest is required to regulate the resonances associated with bound-state formation. Second, we provide the elastic, transfer, and viscosity cross sections computed from the tabulated phase shifts. The cross sections include partial waves up to $\ell = 100$, 99, and 98 for elastic, transfer, and viscosity scattering, respectively. All cross sections are normalized to the s-wave unitarity cross section, $\sigma^{\mathrm{U}}_{\ell=0} = 4\pi/k^2$. These normalized cross sections can be used directly without rerunning the phase-shift calculation or recomputing the cross sections from the phase-shift data.

The phase shifts were computed with `VPSSolve` using a relative convergence tolerance of 10^{-3} on θ_ℓ and an initial working precision of 32 digits, increased automatically where required. Throughout the tabulated parameter space, the truncation error of the cross sections is below 1% [1]. Linear interpolation between grid points introduces additional errors in regions with narrow resonances.

The organization and naming of the corresponding data files are described in the following section.

4.3 Directory structure

All precomputed data are contained within the `./data` directory. The attractive and repulsive data are stored in the directories `AttractiveYukawa` and `RepulsiveYukawa`, with files labeled by `A` and `R`, respectively. Within each, the phase shifts, Sommerfeld factors, cross sections, and partial-wave truncation data are organized into separate subdirectories, as illustrated in Fig. 1.

The parameter grid used for the tabulated quantities is provided in `zetaVSxi_grid.txt`. Within each potential directory, the `PhaseShift` and `Sommerfeld` directories contain the partial-wave phase shifts and Sommerfeld factors, respectively, with one file for each

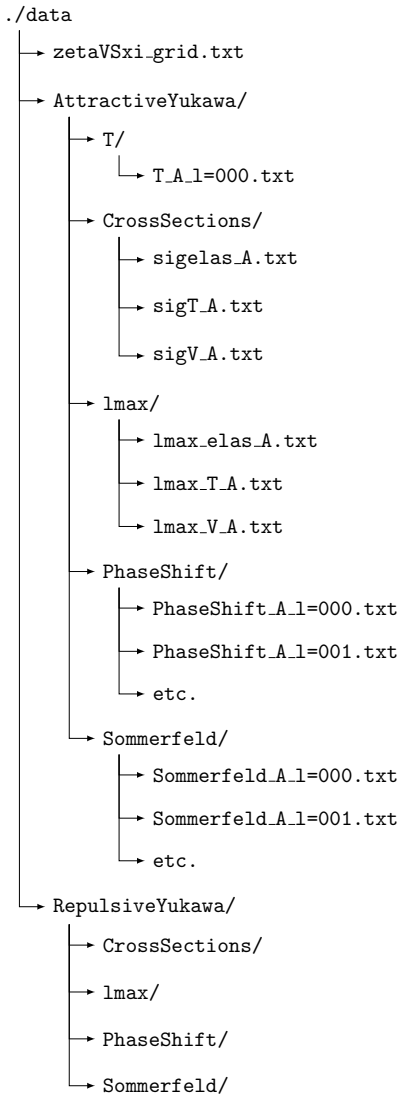


Figure 1: Directory structure of the data files. The subdirectories of `RepulsiveYukawa/` follow those of `AttractiveYukawa/`, with `A` replaced by `R` in the file names. No `T/` directory is provided for the repulsive potential.

value of ℓ. The `CrossSections` directory contains the corresponding elastic, transfer, and viscosity cross sections. The `lmax` directory contains, for each cross section, the smallest $\ell_{\max}$ beyond which the geometric-tail estimate of the relative truncation error remains below 1% [1]. For the attractive potential, the `T` directory additionally contains the tabulated $T(k; k_\diamond)$ data.

4.4 Data file format

All `.txt` files are exported as plain-text tables. The file `zetaVSxi_grid.txt` contains the *complete* $(\xi, |\zeta|)$ parameter grid, with each row specifying one point $(\xi, |\zeta|)$. The rows are ordered with $|\zeta|$ running fastest, i.e., all values of $|\zeta|$ for the first value of ξ, followed by all values of $|\zeta|$ for the second value of ξ, and so on.

The data files contain a single column of values in the same order, but *only* for the points included in the range specified in Section 4.1, i.e., with $\xi \leq 10\,|\zeta|$, the boundary included. Points with $10\,|\zeta| < \xi$ are skipped, so the entries in the data files do not correspond directly to the row numbers in `zetaVSxi_grid.txt`. This convention is used for the phase shifts, Sommerfeld factors, cross sections, $\ell_{\max}$ data, and $T(k; k_\diamond)$.

4.5 Importing the data

Specialized modules are provided to facilitate the import of the key tabulated quantities onto the parameter grid. The `importGrid` module reconstructs the $(\xi, |\zeta|)$ grid from `zetaVSxi_grid.txt`, while an internal subroutine handles the mapping of the data onto the grid and properly accounts for the excluded region.[1] Dedicated import modules are provided for the phase shifts, Sommerfeld factors, and $T(k; k_\diamond)$ data. For example, `importSommerfeldData` facilitates the import of Sommerfeld factors for selected potential types and partial waves.

Example: Importing grid and Sommerfeld factors

`YukawaVPS.nb`:

```
grid = importGrid[];

XIs = grid[[All, 1, 1]];
ZETAs = grid[[1, All, 2]];

Ss = importSommerfeldData[grid, "Types" -> {"A"}, "lValues" -> {0, 1, 2}];

{S0, S1, S2} = {Ss[{"A", 0}], Ss[{"A", 1}], Ss[{"A", 2}]}
```

Figure 2 shows the s-wave Sommerfeld factor for an attractive Yukawa potential over the parameter range defined in Section 4.1.

4.6 Interpolation

Interpolation functions over the provided grid can be constructed using the `makeInterp` module, which is included in both `YukawaVPS.nb` and `CrossSectionsVPS.nb`. Given the parameter grid and corresponding data array, the module returns an interpolation function of $(\xi, |\zeta|)$. The interpolation is performed in $\log_{10}\xi$ and $\log_{10}|\zeta|$, with linear interpolation used by default. Since the interpolation is performed on $\log_{10}$ of the data,

[1] The `importDataOnGrid` subroutine performs this mapping.

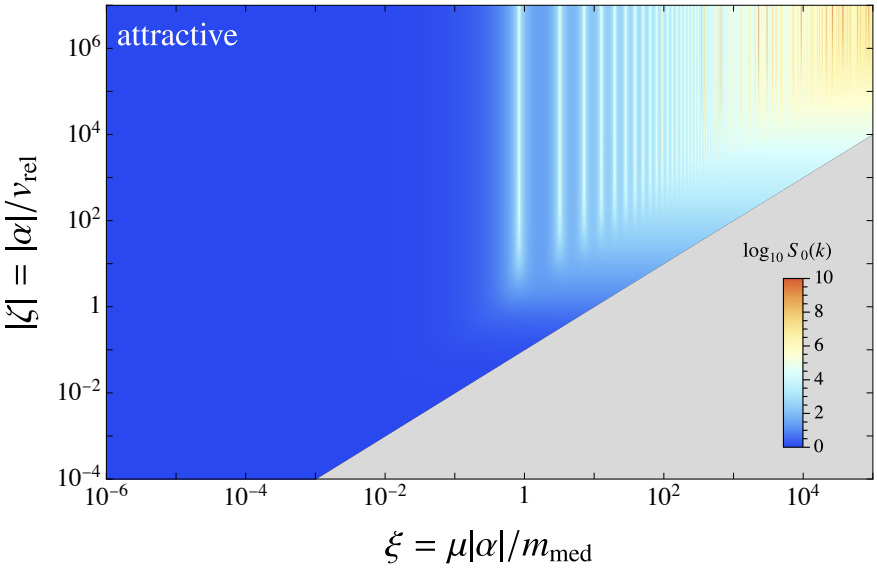


Figure 2: The s-wave Sommerfeld factor for the attractive Yukawa potential over the tabulated $(\xi, |\zeta|)$ parameter space.

`makeInterp` is intended for positive quantities, such as Sommerfeld factors and cross sections. The interpolation order can be specified as an optional third argument. Points outside the provided parameter range or within the excluded region $10\,|\zeta| < \xi$ return `Indeterminate`.

Example: Importing cross sections and building interpolation functions

`CrossSectionsVPS.nb`:

```
grid = importGrid[];

(* Data import *)

SIGMAelasA = importCSLMAXData[grid, "A", "elas", "Data" -> "sigma"];
SIGMATA = importCSLMAXData[grid, "A", "T", "Data" -> "sigma"];
SIGMAVA = importCSLMAXData[grid, "A", "V", "Data" -> "sigma"];

(* Construct interpolation functions *)

SIGMAelasAInterp = makeInterp[grid, SIGMAelasA];
SIGMATAInterp = makeInterp[grid, SIGMATA];
SIGMAVAInterp = makeInterp[grid, SIGMAVA];
```

5 Citation and license

Publications making use of the code or data provided in this repository should cite [1], which describes the methods and calculations implemented here. The code, data, and documentation in this repository are distributed under the MIT License. See the accompanying `LICENSE` file for details.

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